

# Architectural Design, Formal Specification, and Production Implementation of an Elastic Multi-Tier Infrastructure on Amazon Web Services

*A Comprehensive Technical Monograph & Master's Thesis on High Availability, Network Isolation, Zero-Trust Security, and Dynamic Database Interconnectivity for Enterprise Cloud Systems*

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**Author:** Tarra Someswararao | **AWS System Account ID:** 595028889753 (us-east-1)

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## Abstract

Modern cloud computing paradigms demand highly scalable, fault-tolerant, and securely isolated infrastructure topologies to support mission-critical enterprise workloads. This thesis presents a definitive, end-to-end architectural framework and operational implementation of an enterprise-grade 3-Tier Web Architecture hosted on Amazon Web Services (AWS). The system is engineered from foundational networking primitives up to an active production state, deploying a dynamic database-driven web application titled 'Mindcircuit Book Store'.

The architectural lifecycle spans 34 distinct operational steps grouped into four primary phases: (1) Base VPC Network Infrastructure Provisioning, (2) Layered Security & Traffic Ingress Management, (3) Automated Compute Fleet & Database Provisioning via Immutable Golden AMIs and Auto Scaling Groups, and (4) Abstracted Private DNS Interconnectivity and End-to-End Dynamic System Validation. Through rigorous isolation of public web ingress, private application compute, and non-routable relational database subnets across multiple Availability Zones, the architecture achieves zero public exposure of database endpoints while guaranteeing automated self-healing and horizontal elasticity under load.

## Key Architectural Contributions

- **Formal 34-Step Lifecycle Specification:** Complete procedural blueprint detailing exact subnet allocations, route table associations, security group rules, load balancer target groups, and Route 53 DNS mappings.
- **Zero-Trust Network Isolation:** Network Access Control Lists (NACLs) and stateful Security Group chaining restricting lateral movement between tiers.
- **Abstracted Internal Service Discovery:** Deployment of Route 53 Private Hosted Zones (rds.com) mapping human-readable CNAME records (book.rds.com) to dynamic RDS endpoints, eliminating hardcoded infrastructure strings.
- **Automated Elastic Compute Lifecycle:** Immutable image creation pipelines (Golden AMIs) backing Auto Scaling Launch Templates to maintain identical state across dynamic scale-out events.
- **Empirical Verification:** End-to-end live testing validating full CRUD operations from browser clients (virat.rebel7781.xyz) down to persistent MySQL database tables.

# Chapter 1: Theoretical Background & Architectural Requirements

## 1.1 Introduction to Modern Cloud Architecture

Enterprise applications transitioning to public cloud environments must adhere to strict principles of resilience, scalability, and defense-in-depth security. Single-tier or monolithic deployments suffer from high blast radiuses, where a failure in web processing or database query handling crashes the entire application. The 3-Tier Architecture decouples applications into three discrete, independently scalable layers: Presentation Tier (Frontend), Application Tier (Backend Business Logic), and Data Tier (Relational Storage).

## 1.2 Architectural Requirements & Design Constraints

| Requirement Metric     | Target Specification                                              | Architectural Mechanism                                                                                  |
|------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| High Availability (HA) | 99.99% Uptime across Availability Zone outages                    | Multi-AZ deployment (us-east-1a, us-east-1b) with redundant ALBs and ASGs.                               |
| Security Isolation     | Zero direct internet accessibility to Database and Backend layers | Private subnets with strictly managed NAT Gateway egress and chained Security Groups.                    |
| Horizontal Scalability | Automated capacity expansion under CPU/Memory spikes              | EC2 Auto Scaling Groups utilizing Launch Templates backed by Golden AMIs.                                |
| Domain Abstraction     | Zero hardcoded IP addresses or AWS-generated DNS in codebases     | Public Route 53 zones for frontend/API; Private Route 53 hosted zones (rds.com) for internal DB routing. |
| Data Integrity         | ACID-compliant storage with multi-AZ failover capability          | Amazon RDS MySQL instance isolated in a dedicated DB Subnet Group (project-3tier-sn-group).              |

# Chapter 2: Phase 1 — Base Network Infrastructure Provisioning

## 2.1 Production VPC Allocation

The foundational step in building an enterprise AWS cloud presence is establishing an isolated Virtual Private Cloud (VPC). The production VPC, named **3tier-vpc**, is created with an IPv4 CIDR block of **10.20.0.0/16**. DNS Hostnames and DNS Resolution options are explicitly enabled.

## 2.2 Internet Gateway Creation and VPC Attachment

To enable communication between resources in the VPC and the public internet, an Internet Gateway (IGW) named **3tier-igw** is created and explicitly attached to 3tier-vpc.

## 2.3 Multi-AZ Subnet Partitioning

To ensure high availability and physical fault domain isolation, 8 subnets are created across Availability Zones us-east-1a and us-east-1b:

| Subnet Name | CIDR Block   | AZ         | Tier Classification | Route Table Target                        |
|-------------|--------------|------------|---------------------|-------------------------------------------|
| pub-sn-1a   | 10.20.1.0/24 | us-east-1a | Public (ALB / NAT)  | Public Route Table (3tier-pub-rt -> IGW)  |
| pub-sn-2b   | 10.20.2.0/24 | us-east-1b | Public (ALB / NAT)  | Public Route Table (3tier-pub-rt -> IGW)  |
| pvt-sn-3a   | 10.20.3.0/24 | us-east-1a | Private Application | Private Route Table (3tier-pvt-rt -> NAT) |
| pvt-sn-4b   | 10.20.4.0/24 | us-east-1b | Private Application | Private Route Table (3tier-pvt-rt -> NAT) |
| pvt-sn-5a   | 10.20.5.0/24 | us-east-1a | Private Database    | Private Route Table (Local Only)          |
| pvt-sn-6b   | 10.20.6.0/24 | us-east-1b | Private Database    | Private Route Table (Local Only)          |
| pvt-sn-7a   | 10.20.7.0/24 | us-east-1a | Private Auxiliary   | Private Route Table (Local Only)          |
| pvt-sn-8b   | 10.20.8.0/24 | us-east-1b | Private Auxiliary   | Private Route Table (Local Only)          |

## Chapter 3: Phase 2 — Security Architecture & Ingress Traffic Management

### 3.1 Layered Security Group Design

Security Groups function as stateful virtual firewalls at the ENI level. Security groups are chained such that each tier accepts traffic exclusively from the tier directly above it (frontend-alb-sg -> frontend-ec2-sg -> backend-alb-sg -> backend-ec2-sg -> db-sg).

### 3.2 Route 53 & Load Balancer Infrastructure

**Frontend ALB** routes virat.rebel7781.xyz to frontend-TG. **Backend ALB** routes api.rebel7781.xyz to backend-TG. Private Hosted Zone **rds.com** routes CNAME **book.rds.com** to Amazon RDS MySQL instance **somesh-db-1**.

## Chapter 4: Phase 3 — Database, Staging Compute, & Auto Scaling

### 4.1 Amazon RDS MySQL Provisioning

Amazon RDS MySQL instance **somesh-db-1** (MySQL 8.0, db.t3.micro) is isolated in DB Subnet Group **project-3tier-sn-group**.

### 4.2 Auto Scaling & Golden AMIs

Immutable Golden AMIs **frontend-AMI** (ami-0e826fcb0c13a348) and **backend-AMI** (ami-0cf2ba10137800b5a) back Launch Templates **frontend-LT** and **backend-LT** powering **FE-ASG** and **BE-ASG**.

## Chapter 5: Phase 4 — Private DNS Mapping & Dynamic System Validation

End-to-end system verification on <https://virat.rebel7781.xyz> confirms full dynamic database connectivity and CRUD operations across all three tiers.